HW2

RJ Cass

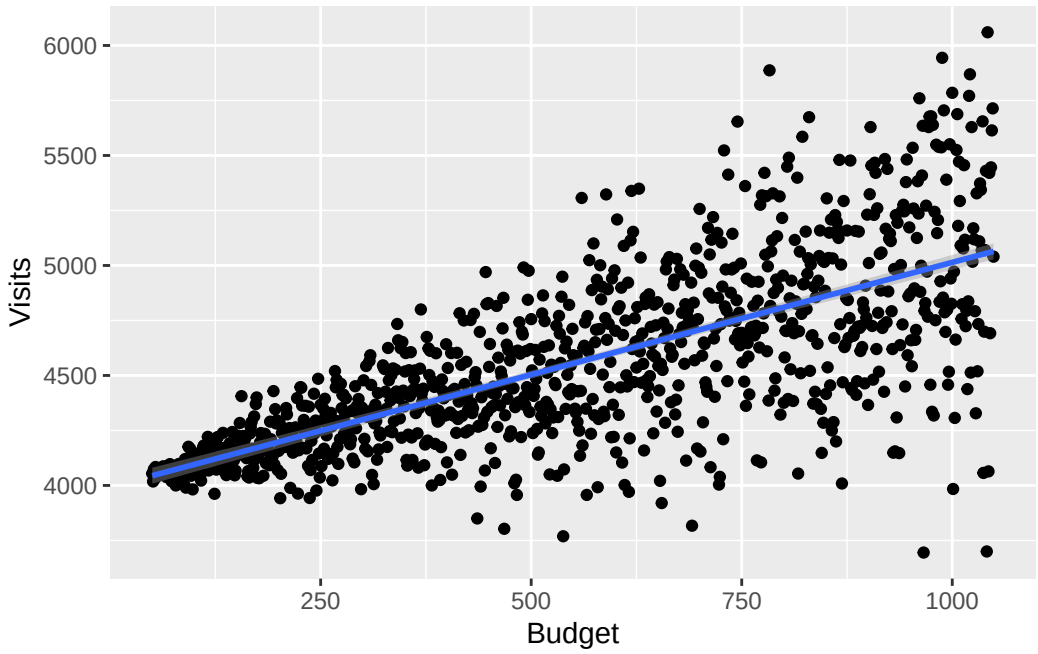
```
suppressPackageStartupMessages(library('ggplot2'))
suppressPackageStartupMessages(library('patchwork'))
suppressPackageStartupMessages(library('gamlss'))
suppressPackageStartupMessages(library('lmtest'))
suppressPackageStartupMessages(library('dplyr'))
web <- read.csv('website.csv')
web$AdType <- as.factor(web$AdType)
```

HW 2.1

(a)

```
p11 <- ggplot(web, aes(x = Budget, y = Visits)) +
    geom_point() +
    geom_smooth(method = "lm", se = TRUE)
p11
```

```
`geom_smooth()` using formula = 'y ~ x'
```

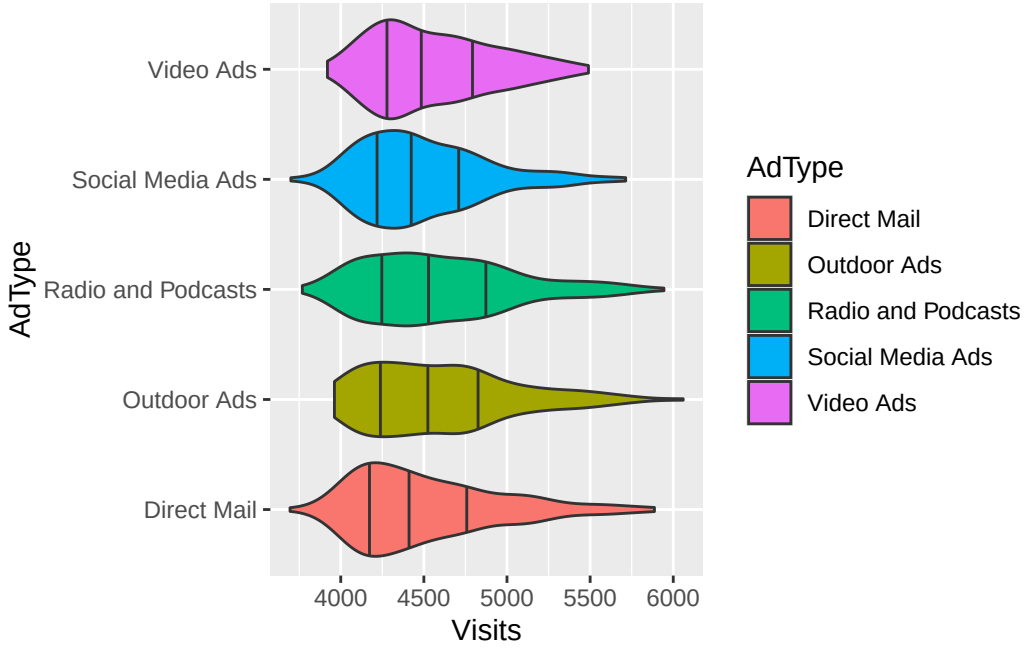


(b)

```
p12 <- ggplot(web, aes(x = Visits, y = AdType, fill = AdType)) +
    geom_violin(draw_quantiles=c(0.25, .5, .75))
```

```
Warning: The `draw_quantiles` argument of `geom_violin()` is deprecated as of ggplot2
4.0.0.
i Please use the `quantiles.linetype` argument instead.
```

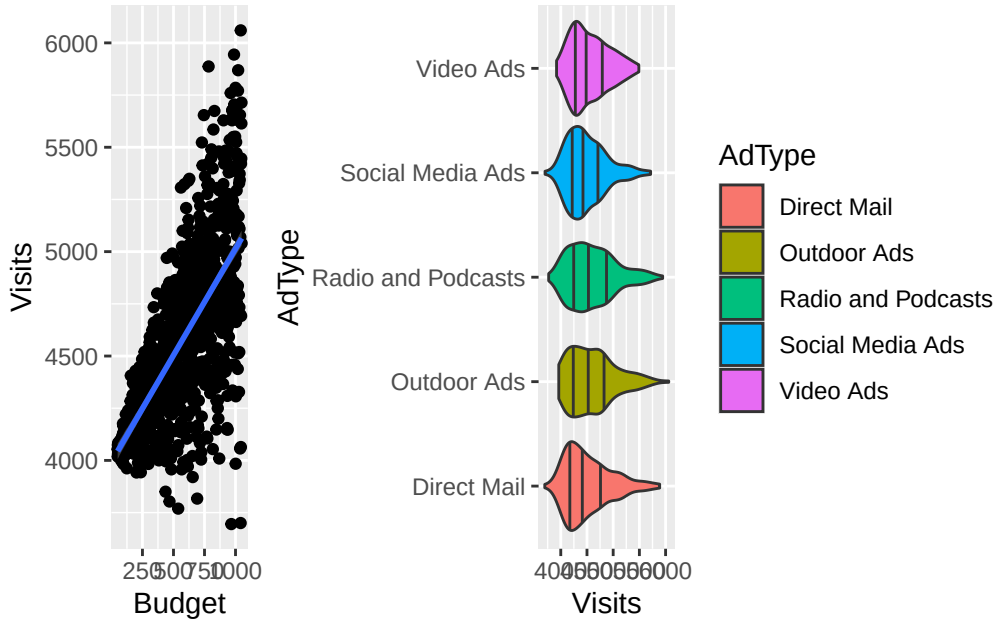
```
p12
```



Plotted together

```
p11 + p12
```

```
`geom_smooth()` using formula = 'y ~ x'
```



(c)

```
mod_budget <- lm(Visits ~ Budget, data = web)
mod_adtype <- lm(Visits ~ AdType, data = web)
mod_full <- lm(Visits ~ Budget + AdType, data = web)

summary(mod_full)
```

```
Call:
lm(formula = Visits ~ Budget + AdType, data = web)

Residuals:
     Min       1Q   Median       3Q      Max
-1313.54  -143.98     3.61   140.65  1095.48

Coefficients:
                          Estimate Std. Error t value Pr(>|t|)
(Intercept)             3995.4368    26.0962 153.104   <2e-16 ***
Budget                     1.0167     0.0322  31.571   <2e-16 ***
AdTypeOutdoor Ads         17.6226    28.9569   0.609    0.543
AdTypeRadio and Podcasts  31.7844    29.0028   1.096    0.273
AdTypeSocial Media Ads   -40.2877    29.3655  -1.372    0.170
AdTypeVideo Ads          -10.3677    28.7366  -0.361    0.718
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 293 on 994 degrees of freedom
Multiple R-squared:  0.506, Adjusted R-squared:  0.5035
F-statistic: 203.6 on 5 and 994 DF,  p-value: < 2.2e-16
```

```
anova(mod_budget, mod_full)
```

```
Analysis of Variance Table

Model 1: Visits ~ Budget
Model 2: Visits ~ Budget + AdType
  Res.Df      RSS Df Sum of Sq     F Pr(>F)
1    998 85928668
2    994 85343805  4    584863 1.703 0.1471
```

```r
anova(mod_adtype, mod_full)
```

```
Analysis of Variance Table

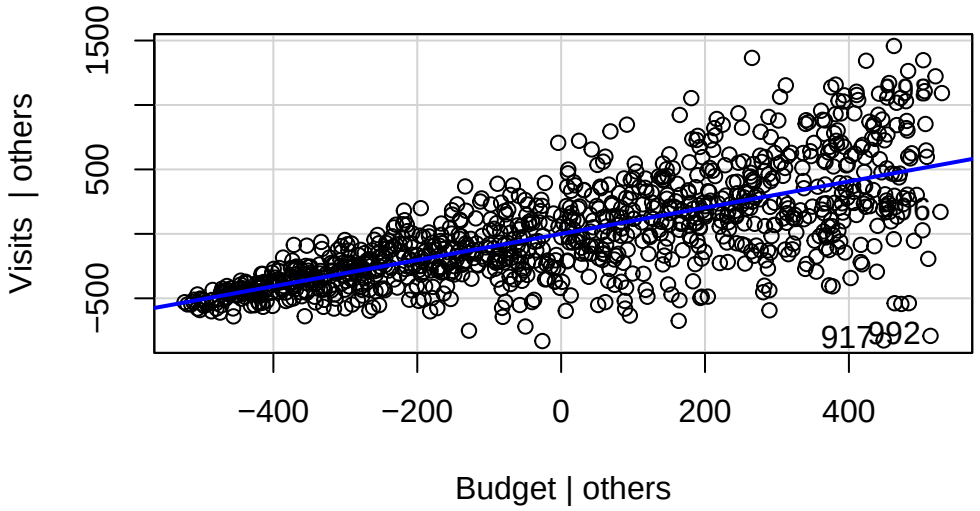
Model 1: Visits ~ AdType
Model 2: Visits ~ Budget + AdType
  Res.Df       RSS Df Sum of Sq     F    Pr(>F)
1    995 170919701
2    994  85343805  1  85575896 996.7 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

AdType does appear to be significant. Budget does not appear to be significant.

(d)

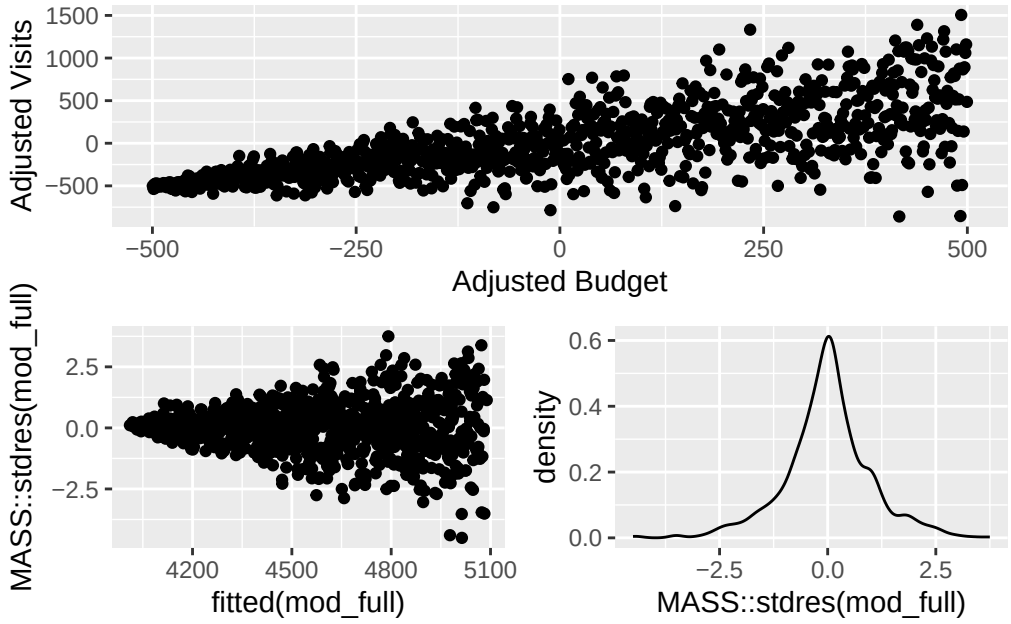
Linearity

```r
car::avPlots(mod_full, terms = ~Budget)
```



```r
p141 <- ggplot(web, aes(x = (Budget - mean(Budget)), y = (Visits - mean(Visits)))) +
    geom_point() +
    labs(x = "Adjusted Budget", y = "Adjusted Visits")
p142 <- ggplot(web, aes(x = fitted(mod_full), y = MASS::stdres(mod_full))) +
    geom_point()
p143 <- ggplot(web, aes(x = MASS::stdres(mod_full))) +
    geom_density()

p141 / (p142 + p143)
```



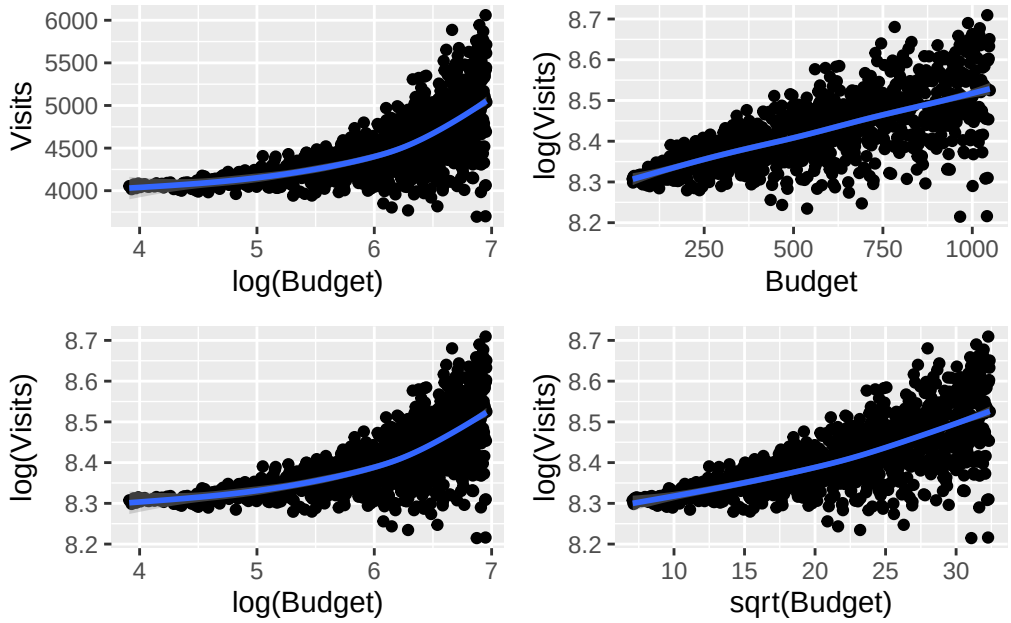
(e)

```r
p151 <- ggplot(web, aes(x = log(Budget), y = Visits)) +
    geom_point() +
    geom_smooth()
p152 <- ggplot(web, aes(x = Budget, y = log(Visits))) +
    geom_point() +
    geom_smooth()
p153 <- ggplot(web, aes(x = log(Budget), y = log(Visits))) +
    geom_point() +
    geom_smooth()
p154 <- ggplot(web, aes(x = sqrt(Budget), y = log(Visits))) +
```

```
    geom_point() +
    geom_smooth()


(p151 + p152) / (p153 + p154)
```

```
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
`geom_smooth()` using method = 'gam' and formula = 'y ~ s(x, bs = "cs")'
```



HW 2.2

(a)

```
gam_mod <- gamlss(Visits ~ Budget + AdType, sigma.formula = ~ Budget, data = web)
```

```
GAMLSS-RS iteration 1: Global Deviance = 13740.17
GAMLSS-RS iteration 2: Global Deviance = 13724.39
GAMLSS-RS iteration 3: Global Deviance = 13724.38
GAMLSS-RS iteration 4: Global Deviance = 13724.38
```

Estimates for the βs

```
coef(gam_mod, what = "mu") # Mean model coefficients (betas)
```

```
             (Intercept)                  Budget        AdTypeOutdoor Ads
            3994.3226521               1.0051834               8.1050678
AdTypeRadio and Podcasts   AdTypeSocial Media Ads          AdTypeVideo Ads
              10.5937507               0.3614247              10.5669007
```

Estimates for θ_0 and θ_1

```
coef(gam_mod, what = "sigma") # Variance model coefficients
```

```
(Intercept)      Budget
4.336147892 0.002014751
```

(b)

I added in a grid to start over to demonstrate the change point where it starts converging to the different peaks.

```
multi_modal <- function(x) {
    # Parameters for the first peak
    mean1 <- -2
    sd1 <- 1
    peak1 <- dnorm(x, mean = mean1, sd = sd1)

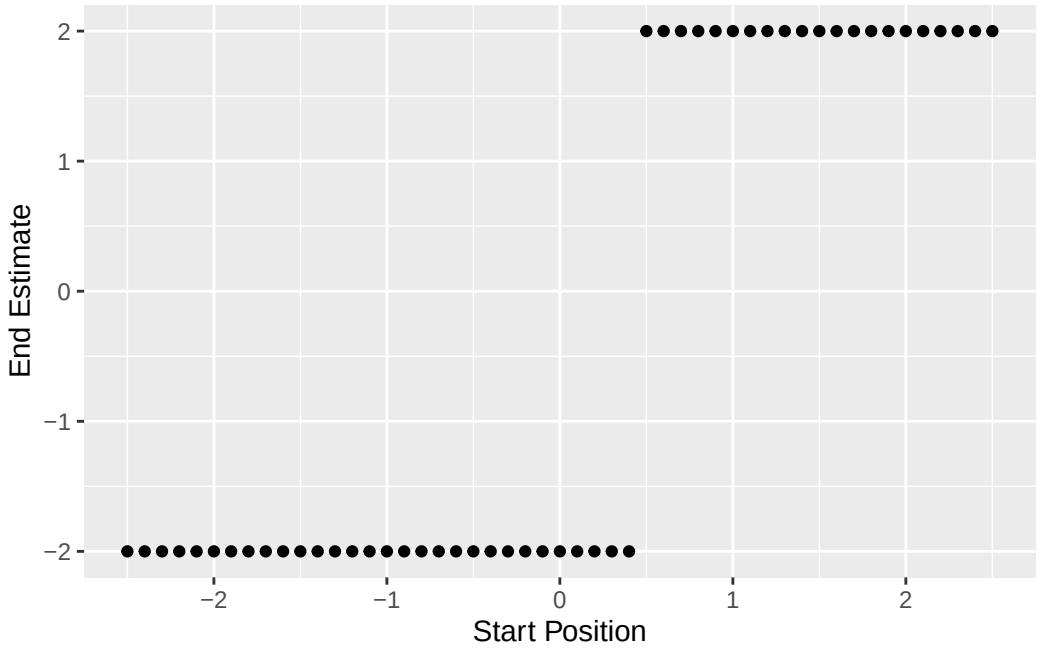
    # Parameters for the second peak
    mean2 <- 2
    sd2 <- .5
    peak2 <- dnorm(x, mean = mean2, sd = sd2)

    # Sum of both peaks
    peak1 + peak2
}

start <- seq(-2.5, 2.5, by = .1)
end <- unlist(lapply(start, \(x) optim(x, multi_modal, method = "L-BFGS-B", control = list(fn

ggplot(NULL, aes(x = start, y = end)) +
```

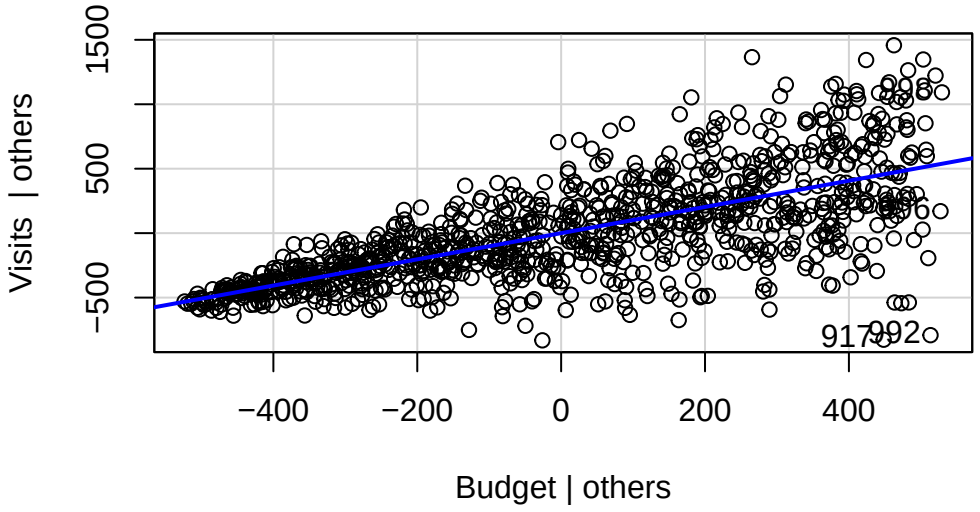
```
    geom_point() +
    labs(x = "Start Position", y = "End Estimate")
```



HW 2.3

(a)

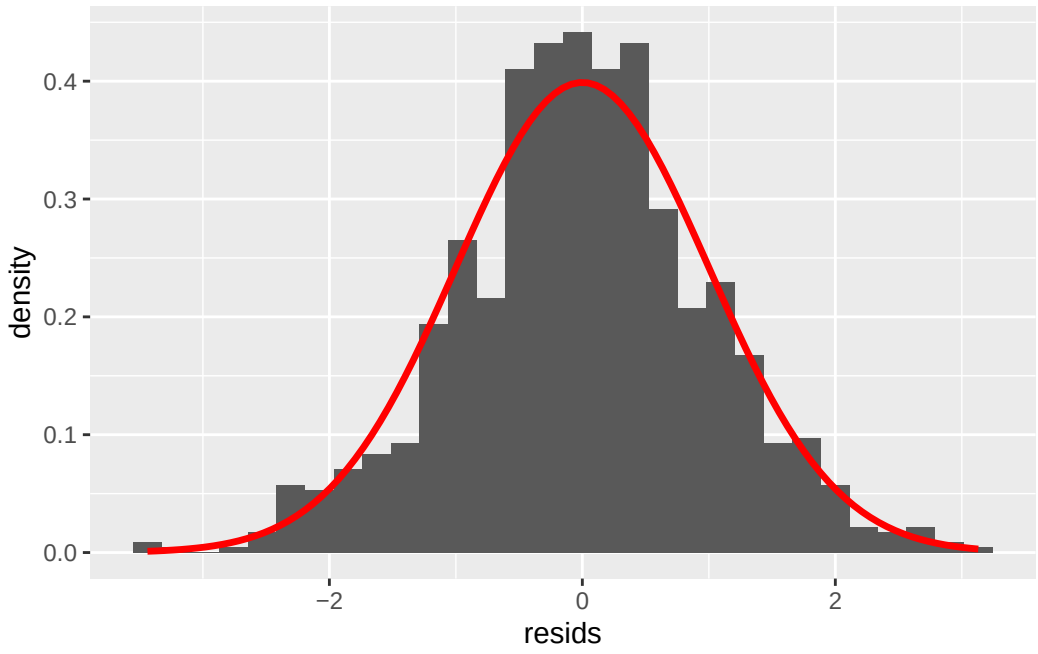
```
car::avPlots(mod_full, terms = ~Budget)
```



(b)

```
resids <- residuals(gam_mod)
resids <- as.data.frame(x = resids)
ggplot(data = resids, aes(x=resids)) +
    geom_histogram(aes(y = after_stat(density))) +
    stat_function(fun = dnorm, color = "red", linewidth = 1.2)
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
ks.test(resids$resids, "pnorm")
```

```
    Asymptotic one-sample Kolmogorov-Smirnov test

data:  resids$resids
D = 0.031495, p-value = 0.2744
alternative hypothesis: two-sided
```

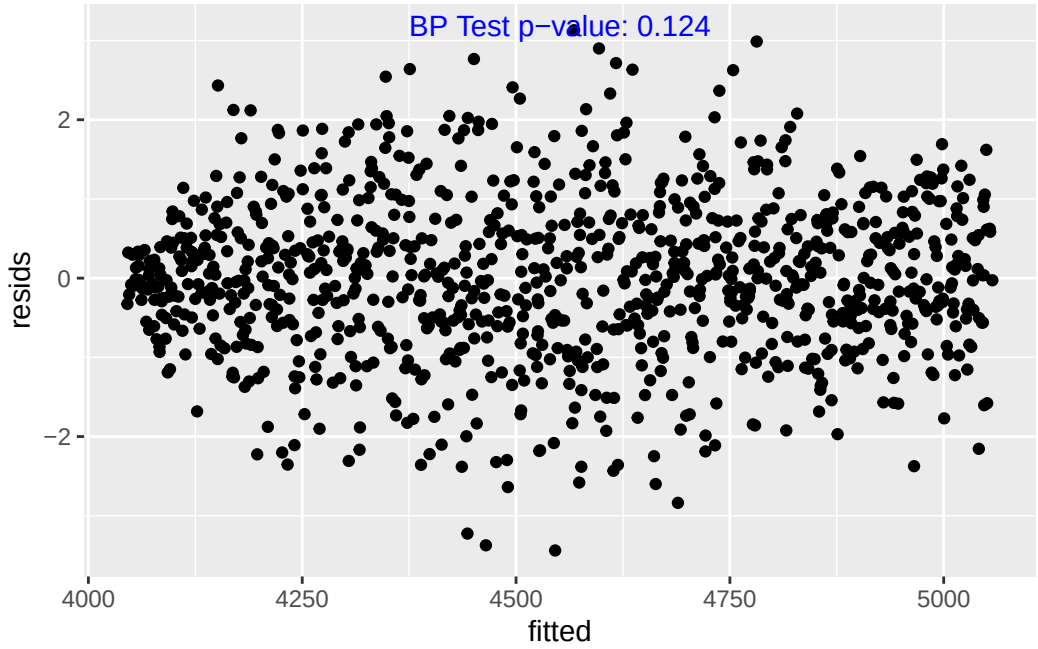
(c)

```
res_squared <- residuals(gam_mod)^2
aux_mod <- lm(res_squared ~ Budget + AdType, data = web)

bp <- bptest(aux_mod)
p_val <- bp$p.value
p_label <- paste("BP Test p-value:", format.pval(p_val, digits = 3))

fits <- fitted(gam_mod, what = "mu")
ggplot(NULL, aes(x = fits, y = resids$resids)) +
    geom_point() +
```

```
    labs(x = "fitted", y = "resids") +
    annotate("text", x = median(fits), y = Inf, label = p_label, vjust = 1.5, color = "blue")
```



(d)

```
# R-squared-ish formula
cor(web$Visits, fitted(gam_mod, what = "mu"))^2
```

```
[1] 0.5032893
```

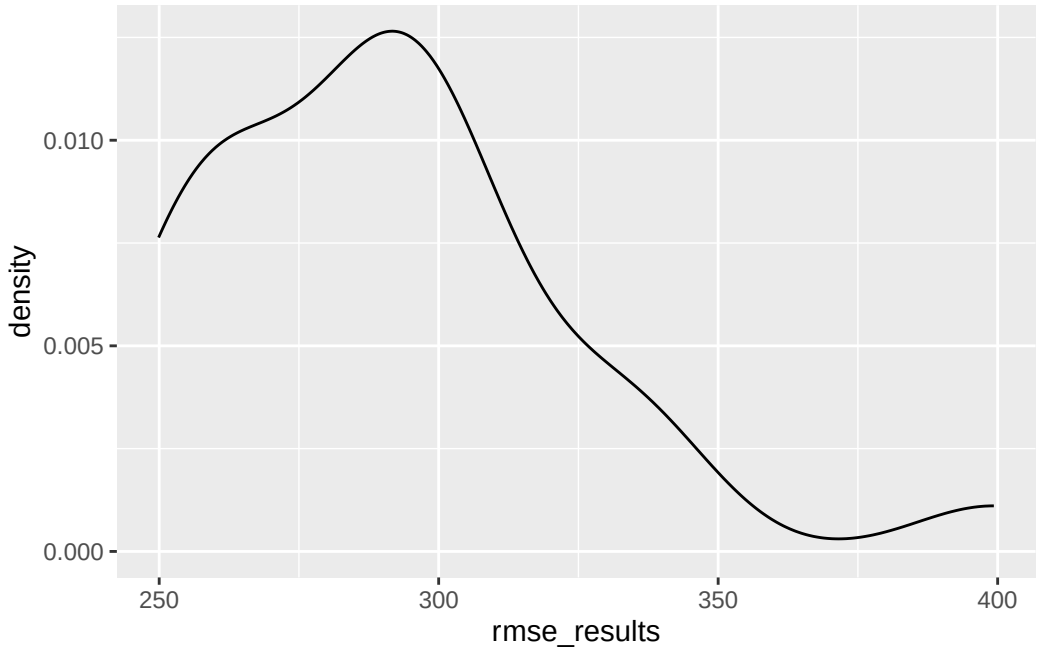
(e)

```
cross_validate <- function(fold_num) {
    train_dt <- web[folds != fold_num, ]
    val_dt <- web[folds == fold_num, ]
    cv_mod <- gamlss(Visits ~ Budget + AdType, sigma.formula = ~ Budget, data = train_dt, tra
    preds <- predict(cv_mod, what = "mu", data = train_dt, newdata = val_dt, type = "response
    raw_resids <- val_dt$Visits - preds
    rmse <- sqrt(mean(raw_resids^2))
```

```
    rmse
}

set.seed(1337)
K <- 25
folds <- rep(1:K, length = nrow(web)) %>% #Rep K folds n times
    sample()
## Using sapply()
rmse_results <- sapply(1:K, FUN=cross_validate)

## Plot the Results
ggplot() +
    geom_density(mapping=aes(x=rmse_results))
```



HW 2.4

(a)

```
mod <- gamlss(formula = Visits ~ Budget + AdType, sigma.formula = ~Budget, data = web)
```

```
GAMLSS-RS iteration 1: Global Deviance = 13740.17
GAMLSS-RS iteration 2: Global Deviance = 13724.39
GAMLSS-RS iteration 3: Global Deviance = 13724.38
GAMLSS-RS iteration 4: Global Deviance = 13724.38
```

```
summary(mod)
```

```
******************************************************************
Family:  c("NO", "Normal")

Call:  gamlss(formula = Visits ~ Budget + AdType, sigma.formula = ~Budget,
    data = web)

Fitting method: RS()

------------------------------------------------------------------
Mu link function:  identity
Mu Coefficients:
                          Estimate Std. Error t value Pr(>|t|)
(Intercept)              3.994e+03  1.260e+01 316.896   <2e-16 ***
Budget                   1.005e+00  2.634e-02  38.157   <2e-16 ***
AdTypeOutdoor Ads        8.105e+00  1.650e+01   0.491    0.623
AdTypeRadio and Podcasts 1.059e+01  1.667e+01   0.635    0.525
AdTypeSocial Media Ads   3.614e-01  1.619e+01   0.022    0.982
AdTypeVideo Ads          1.057e+01  1.709e+01   0.618    0.537
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

------------------------------------------------------------------
Sigma link function:  log
Sigma Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept) 4.336e+00  4.250e-02  102.02   <2e-16 ***
Budget      2.015e-03  4.806e-05   41.92   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

------------------------------------------------------------------
No. of observations in the fit:  1000
Degrees of Freedom for the fit:  8
      Residual Deg. of Freedom:  992
                      at cycle:  4
```

```
Global Deviance:     13724.38
            AIC:     13740.38
            SBC:     13779.65
*******************************************************************
```

Looking at the initial summary output, we see that the p-value for Budget is almost 0 (< 2e-16). Due to this, we conclude that Budget does have a significant effect on the number of visists.

```r
mu_cis <- confint(mod, what = "mu", level = 0.95)

# Convert it into a clean table structure
mu_cis_table <- data.frame(
  Beta_Estimate = coef(mod, what = "mu"),
  "2.5%"       = mu_cis[, 1],
  "97.5%"      = mu_cis[, 2]
)

print(mu_cis_table)
```

```
                        Beta_Estimate        X2.5.      X97.5.
(Intercept)              3994.3226521 3969.6182270 4019.027077
Budget                      1.0051834    0.9535508    1.056816
AdTypeOutdoor Ads           8.1050678  -24.2337316   40.443867
AdTypeRadio and Podcasts   10.5937507  -22.0871968   43.274698
AdTypeSocial Media Ads      0.3614247  -31.3711511   32.094001
AdTypeVideo Ads            10.5669007  -22.9346190   44.068420
```

(b)

```r
mod_no_ad <- gamlss(formula = Visits ~ Budget, sigma.formula = ~Budget, data = web)
```

```
GAMLSS-RS iteration 1: Global Deviance = 13725.55
GAMLSS-RS iteration 2: Global Deviance = 13725.17
GAMLSS-RS iteration 3: Global Deviance = 13725.17
```

```
LR.test(mod_no_ad, mod)
```

```
 Likelihood Ratio Test for nested GAMLSS models.
 (No check whether the models are nested is performed).

       Null model: deviance= 13725.17 with  4 deg. of freedom
 Altenative model: deviance= 13724.38 with  8 deg. of freedom

 LRT = 0.7853942 with 4 deg. of freedom and p-value= 0.9403954
```

Due to the p-value of .94 we conclude that the reduced model performs as well as the full model, implying that AdType does not have a significant effect on the number of visits.

(c)

```
sigma_cis <- confint(mod, what = "sigma", level = 0.95)
sigma_cis
```

```
                       2.5 %      97.5 %
sigma.(Intercept) 4.25284377 4.419452012
sigma.Budget      0.00192056 0.002108943
```

(d)

```
budget_vals <- min(web$Budget):max(web$Budget)
pred_vals <- data.frame(Budget = budget_vals)

budget_preds <- predictAll(mod_no_ad, data = web, newdata = pred_vals, type = "response")

lower_90 <- qNO(0.05, mu = budget_preds$mu, sigma = budget_preds$sigma)
upper_90 <- qNO(0.95, mu = budget_preds$mu, sigma = budget_preds$sigma)

# 3. Combine them into a clean data frame with your predictions
budget_intervals <- data.frame(
    budget = budget_vals,
    fit    = budget_preds$mu,
    lower = lower_90,
```

```
    upper = upper_90
)

ggplot(data = budget_intervals, aes(x = budget, y = fit)) +
    geom_point(data = web, aes(x = Budget, y = Visits), color = "black") +
    geom_line(color = "red", lwd = 2) +
    geom_line(aes(y = lower), color = "blue", lwd = 2) +
    geom_line(aes(y = upper), color = "blue", lwd = 2) +
    labs(x = "Budget", y = "Visits") +
    theme_minimal()
```

